Course Code: CSE-1213
Course Title: Discrete Mathematics
3 hours in a week, 3.00 Cr.

Course Teachers:

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Course Code: CSE-1211
Course Title: Discrete Mathematics
3 hours in a week, 3.00 Cr.

- **Mathematical Logic:** propositional calculus and predicate calculus, Mathematical reasoning and proof techniques**.**
- **Set theory**: sets, relations, partially ordered sets, functions.
- **Counting:** permutations, combinations, principles of inclusion and exclusion; Discrete Probability; Recurrence relations and recursive algorithms; Growth of functions.
- **Graph theory:** graphs, paths, trees, cycles; Algebraic structures: rings and groups: Groyas Semi group, Monoid Groups, Abelian group, properties of groups, Permutation Groups, Subgroups, Cyclic Group.

Reference Book(s):

1. Discrete Mathematics & Its Applications- Kenneth H Rosen
2. Discrete Mathematics - Seymour Lipschutz, M. Lipson, Tata McGraw-Hill

Propositional Logic and Proof

What is Discrete Mathematics?

From http://dictionary.oed.com:

Discrete

1. a. Separate, detached from others, individually distinct. Opposed to *continuous*.

2. a. Consisting of distinct or individual parts; discontinuous.

According to writer **Kenneth H. Rosen-**

Discrete mathematics is the part of mathematics devoted to the study of discrete objects.

Kind of problems solving using discrete mathematics

- *What is the probability of winning a lottery?*
- *Is there a link between two computers in a network?*
- *What is the shortest path between two cities using a transportation system?*
- *How can a list of integers be sorted so that the integers are in increasing order?*
- *How many steps are required to do such a sorting?*
- *How can it be proved that a sorting algorithm correctly sorts a list?*
- *How can a circuit that adds two integers be designed?*

Application Area

- Advanced algorithms & data structures
- Programming language compilers & interpreters.
- Computer networks
- Operating systems
- Computer architecture
- Database management systems
- Cryptography
- Error correction codes
- Graphics & animation algorithms, game engines, *etc.*…
- *I.e.*, the whole field!

Books?

Discrete Mathematics

Schema's outline series: Seymour Lipshutz.

Discrete Mathematics and Its Applications

McGraw-Hill: Kenneth H. Rosen

References

Discrete Mathematics and its Applications by *Kenneth H. Rosen*

- Chapter 1 : The Foundation : Logic and Proof

Discrete Mathematics by *Seymour Lipschutz, Marc Lipso*

- Chapter 4 : Logic and Propositional Calculas

Today's Topics

- Basic Propositional Logic
- Variable Based Propositional Logic
- Rules of Inference
- Proof Technique

Proposition

Definition

A proposition is a declarative sentence *(that is, a sentence that declares a fact)* that is either true or false, but not both.

Example

- Washington, D.C., is the capital of the United States of America
- Dhaka is the capital of Bangladesh
- Dhaka is the capital of India
- 8 + 2 = 10
- 4 + 4 = 10

Proposition or Not?

- What time is it? ✗
- Read this carefully. ✗
- $a + 2 = 10$ ✗
- $x + y = z$ ✗

Propositional Variable

Definition

variables that represent propositions, just as letters are used to denote numerical variables.

The conventional letters used for propositional variables are p, q, r, s, ...

Example :

p = Dhaka is the capital of Bangladesh.

Some common terms

The truth value

The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.

$p = Dhaka\ is\ the\ capital\ of\ Bangladesh.$	Truth value : T
$p = Dhaka\ is\ the\ capital\ of\ Pakistan.$	Truth value : F

Some common terms

Propositional Calculus or Propositional Logic.

The area of logic that deals with propositions is called the propositional calculus or propositional logic.

Compound Propositions

New propositions formed from existing propositions using logical operators are called compound propositions.

Introduction to Logical Operators

Basic Operators

1. Negation (NOT)
2. Conjunction (AND)
3. Disjunction (OR)

Logical operators: Not

Definition

A "not" operation switches (negates) the truth value.

Symbol: $\neg$ or ~

- If $p = To\ day\ is\ Monday$.
- $\neg p = ?$

Truth table	
P	$\neg p$
T	F
F	T

Logical operators: And

Definition

An "and" operation is true if both operands are true.

Symbol: $\wedge$

p = "Today is Friday"
q = "Today is my birthday"

$p \wedge q$ = ?

$p \wedge q$ = "Today is Friday and today is my birthday"

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Logical operators: Or

Definition

An "or" operation is true if either operands or both are true.

Symbol: $\vee$

p = *"Today is Friday"*
q = *"Today is my birthday"*

$p \vee q$ = ?

$p \vee q$ = "Today is Friday or today is my birthday"

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

More Logical Operators

More Operators

1. Exclusive OR
2. Conditional
3. Bicondition

Logical operators: Exclusive Or

Definition

Let p and q be propositions. The exclusive or of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

- Symbol: $\oplus$
- Often called XOR
- $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$

$p \oplus q$ = "Today is Friday or today is my birthday, but not both"

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Logical operators: Conditional

Definition

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p, then q." The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.

$p \rightarrow q$ = "If today is Friday, then today is my birthday"

- In the conditional statement $p \rightarrow q$,
- p is called the hypothesis (or antecedent or premise)
- q is called the conclusion (or consequence)

- *p* implies *q*
- If *p*, *q*
- *p* only if *q*
- *p* is sufficient for *q*
- *q* if *p*
- *q* whenever *p*
- *q* is necessary for *p*

Logical operators: Conditional

the antecedent

the consequence

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Let, p = "Maria learns discrete mathematics" and

q = "Maria will find a good job."

Express the statement p→q as a statement in English.

Logical operators: Bi-Conditional

Definition

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition "p if and only if q." The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.

Biconditional statements are also called bi-implications.

- "p is necessary and sufficient for q"
- "if p then q, and conversely"

Logical operators: Bi-Conditional

- True when both has same truth values and otherwise false.

- Alternatively, it means "(if *p* then *q*) and (if *q* then *p*)" ***(p→q)∧ (q→p)***.
- Note that a bi-conditional has the opposite truth values of the exclusive or

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- Let *p* = "You take this class" and *q* = "You get a grade"
- $p \leftrightarrow q$ = "You take this class if and only if you get a grade"
- Alternatively, it means "If you take this class, then you get a grade and if you get a grade then you take (took) this class"

Truth Tables of Compound Propositions

Example

Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$.

TABLE 7 The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Boolean operators summary

		and	or	xor	conditional	Bi-conditional
p	q	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	F	T	T
T	F	F	T	T	F	F
F	T	F	T	T	T	F
F	F	F	F	F	T	T

Precedence of operators

- Just as in algebra, operators have precedence

  4+3*2 = ?

  - 4+3*2 = 4+(3*2)

  - 4+3*2 = (4+3)*2

Precedence of operators

- $p \vee q \wedge \neg r \rightarrow s \leftrightarrow t$ = ?

$=(p \vee (q \wedge (\neg r)) \rightarrow s) \leftrightarrow (t)$

- First three are the most important
- Not is *always* performed before any other operation

Operators	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Logic and Bit Operations

Truth Value	*Bit*
T	1
F	0

TABLE 9 Table for the Bit Operators *OR*, *AND*, and *XOR*.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

```
 01 1011 0110
 11 0001 1101
 ------------
 11 1011 1111   bitwise OR
 01 0001 0100   bitwise AND
 10 1010 1011   bitwise XOR
```

Applications of Propositional Logic

- Translating English Sentences
- System Specifications
- Boolean Searches
- Logic Puzzles
- Logic Circuits

Translating English Sentences

Example

p = "It is below freezing"
q = "It is snowing"

It is below freezing and it is snowing	$p \wedge q$
It is below freezing but not snowing	$p \wedge \neg q$
If it is below freezing, it is also snowing	$p \rightarrow q$
It is either below freezing or it is snowing, but it is not snowing if it is below freezing	$(p \vee q) \wedge (p \rightarrow \neg q)$
That it is below freezing is necessary and sufficient for it to be snowing	$p \leftrightarrow q$

Translating English Sentences

Problem: How can this English sentence be translated into a logical expression?

"You can access the Internet from campus only if you are a computer science major or you are not a freshman."

Solve:

Let, a = *"You can access the Internet from campus,"*

c = "You are a computer science major," and

d = "You are a freshman,"

Then the statement can be represented as $a \rightarrow (c \vee \neg f)$.

Translating English Sentences

Problem: How can this English sentence be translated into a logical expression?

"You **can access the Internet** from campus **only if** you are **a computer science major or** you are **not a freshman**."

Solve:

Let, a = *"You can access the Internet from campus,"*

c = "You are a computer science major," and

f = "You are a freshman,"

Then the statement can be represented as $a \rightarrow (c \vee \neg f)$.

Translating English Sentences

How can this English sentence be translated into a logical expression?

"You can not ride the roller coaster if you are under 4 feet tall and you are older than 16 years old."

Solution: Let ***q, r, and s*** *represent "You can ride the roller coaster," "You are under 4 feet tall," and "You are older than 16 years old," respectively. Then the sentence can be translated to* $(r \wedge s) \rightarrow \neg q$.

Exercise 9

Let p:You drive over 65 miles per hour.
q: You get a Speeding ticket.

Write these propositions using *p* and *q* and logical connectives

a) You do not drive over 65 miles per hour.

b) You drive over 65 miles per hour, but you do not get a speeding ticket.

c) You will get a speeding ticket if you drive over 65 miles per hour.

d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.

e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.

f) You get a speeding ticket, but you do not drive over 65 miles per hour.

g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

Propositional Equivalences

Back to the ¬Future

Propositional Calculus or Propositional Logic.

The area of logic that deals with propositions is called the propositional calculus or propositional logic.

Compound Propositions

New propositions formed from existing propositions using logical operators are called compound propositions.

Some Common Terms

Tautology

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a tautology.

Contradiction

A compound proposition that is always false is called a contradiction.

Contingency

A compound proposition that is neither a tautology nor a contradiction is called a contingency.

Example

Tautology

$p \vee \neg p$

Contradiction

$p \wedge \neg p$

TABLE 1 Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Contingency

$p \rightarrow \neg p$

Tautology

Demonstrate that

$$[\neg p \wedge (p \vee q)] \rightarrow q$$

is a tautology in two ways:

1. Using a truth table – show that $[\neg p \wedge (p \vee q)] \rightarrow q$ is always true
2. Using a proof (will get to this later).

Tautology by truth table

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T				
T	F				
F	T				
F	F				

Tautology by truth table

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F			
T	F	F			
F	T	T			
F	F	T			

Tautology by truth table

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F	T		
T	F	F	T		
F	T	T	T		
F	F	T	F		

Tautology by truth table

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F	T	F	
T	F	F	T	F	
F	T	T	T	T	
F	F	T	F	F	

Tautology by truth table

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	F	F	T

Propositional Equivalences

Definition

The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

- Compound propositions that have the same truth values in all possible cases are called logically equivalent.

Propositional Equivalences : How to Prove?

- Two methods:
  - Using truth tables
    - Not good for long formula
    - In this course, only allowed if specifically stated!
  - Using the logical equivalences
    - The preferred method
- $p \rightarrow q \equiv \neg q \rightarrow \neg p$
- $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Logical Equivalence Using Truth Table

The easiest way to check for logical equivalence is to see if the truth tables of both variants have ***identical last columns***. ($p \rightarrow q \equiv \neg q \rightarrow \neg p$)

p	q	$p \rightarrow q$

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$

Logical Equivalence Using Truth Table

The easiest way to check for logical equivalence is to see if the truth tables of both variants have *identical last columns*:

p	q	$p \rightarrow q$
T	T	**T**
T	F	**F**
F	T	**T**
F	F	**T**

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$

Logical Equivalence Using Truth Table

The easiest way to check for logical equivalence is to see if the truth tables of both variants have *identical last columns*:

p	q	$p \rightarrow q$
T	T	**T**
T	F	**F**
F	T	**T**
F	F	**T**

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T			
T	F			
F	T			
F	F			

Logical Equivalence Using Truth Table

The easiest way to check for logical equivalence is to see if the truth tables of both variants have *identical last columns*:

p	q	$p \rightarrow q$
T	T	**T**
T	F	**F**
F	T	**T**
F	F	**T**

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	F		
T	F	T		
F	T	F		
F	F	T		

Logical Equivalence Using Truth Table

The easiest way to check for logical equivalence is to see if the truth tables of both variants have *identical last columns*:

p	q	$p \rightarrow q$
T	T	**T**
T	F	**F**
F	T	**T**
F	F	**T**

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	F	F	
T	F	T	F	
F	T	F	T	
F	F	T	T	

Logical Equivalence Using Truth Table

The easiest way to check for logical equivalence is to see if the truth tables of both variants have *identical last columns*:

p	q	$p \rightarrow q$
T	T	**T**
T	F	**F**
F	T	**T**
F	F	**T**

p	q	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	F	F	**T**
T	F	T	F	**F**
F	T	F	T	**T**
F	F	T	T	**T**

Propositional Equivalences : How to Prove?

- Two methods:
  - Using truth tables
    - Not good for long formula
    - In this course, only allowed if specifically stated!
  - Using the logical equivalences
    - The preferred method
- $p \rightarrow q \equiv \neg q \rightarrow \neg p$
- $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Truth Table Solution

- $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

p	q	r	$p \rightarrow r$	$q \rightarrow r$	$p \wedge q$	$(p \rightarrow r) \vee (q \rightarrow r)$	$(p \wedge q) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	T	T	F	T	T
T	F	F	F	T	F	T	T
F	T	T	T	T	F	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	F	T	T
F	F	F	T	T	F	T	T

Logical Equivalences

$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity Laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination Law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent Laws	$\neg (p \wedge q) \equiv \neg p \vee \neg q$ $\neg (p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$\neg(\neg p) \equiv p$	Double negation law	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative Laws	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation lows
$p \rightarrow q \equiv \neg p \vee q$	Definition of Implication	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$	Definition of Biconditional

Proof using Logical Equivalence

$(p \rightarrow r) \vee (q \rightarrow r)$	
$\equiv (\neg p \vee r) \vee (\neg q \vee r)$	Definition of implication
$\equiv \neg p \vee r \vee \neg q \vee r$	Associative
$\equiv \neg p \vee \neg q \vee r \vee r$	Commutative
$\equiv (\neg p \vee \neg q) \vee (r \vee r)$	Associative
$\equiv \neg (p \wedge q) \vee r$	De Morgan, Idempotent
$\equiv (p \wedge q) \rightarrow r$	Definition of implication

Logical Equivalences : Example

- Show that $(p \wedge q) \rightarrow (p \vee q)$ is a Tautology. (Proof)

$(p \wedge q) \rightarrow (p \vee q)$

$\equiv \neg (p \wedge q) \vee (p \vee q)$ Implication

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ De Morgan

$\equiv (\neg p \vee p) \vee (\neg q \vee q)$ Commutative, Associative

$\equiv T \vee T$ Negation

$\equiv T$ Identity

TASK

- **Use a truth table to verify**

  i) $p \wedge (q \wedge r) \equiv (p \wedge q)\ \ (p \wedge r)$

  ii) $\neg (p \wedge q) \equiv \neg p \vee \neg q$

- **Show that each of these conditional statements are tautology by using truth table.**

i) $\neg p \rightarrow (p \rightarrow q)$ **ii)** $(p \wedge q) \rightarrow (p \rightarrow q)$

Iii) $\neg(p \rightarrow q) \rightarrow p$ **iv)** $\neg(p \rightarrow q) \rightarrow \neg q$

- Show that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent
- Show that $\neg p \leftrightarrow q$ and $p \leftrightarrow \neg q$ are logically equivalent (using truth table)

Predicates (Propositional Function)
and
Quantifiers

Predicates (Propositional Function)

- So far we can represent the statement – “10 > 100” or “10 <= 100”
  - But we can’t represent - “x > 3,” “x = y + 3,” “x + y = z,”
- Can represent – “Niloy knows C programming.”
  - Can’t represent – “X knows C programming.”
  - Can’t represent – “Every student of 3rd batch of BAUET know C programming.”
  - Can’t represent – “At least one student know C programming.”

Predicates : “x is greater than 3”

“x is greater than 3” has two parts

- “**x** is greater than 3” – the variable x, is the subject of the statement.
- “x **is greater than 3**” – the predicate, refers to a property that the **subject** of the statement can have.

- We can denote the statement “x is greater than 3” by P(x),
  - Where P denotes the predicate “is greater than 3” and
  - x is the variable.

Predicates : “x is greater than 3”

- We can denote the statement “x is greater than 3” by P(x),
  - Where P denotes the predicate “is greater than 3” and
  - x is the variable.
- The statement P(x) is also said to be the value of the propositional function P at x.

- Once a value has been assigned to the variable x, the statement P(x) becomes a proposition and has a truth value.

Q: Let P(x) denote the statement “x > 3.” What are the truth values of P(4) and P(2)?

Predicates : Some Example

Example 1

Let Q(x, y) denote the statement “x = y + 3.” What are the truth values of the propositions Q(1, 2)and Q(3, 0)?

Example 3

Let R(x, y, z) denote the statement “x + y = z.” What are the truth values of the propositions R(1,2,3) and R(0,0,1)?

Quantifiers

What is it?

Quantification is a way to create a proposition from a propositional function (predicate).

Quantification **expresses the extent** to which a **predicate is true** over a **range of elements.**

- English words **all, some, many, none, and few** are used in quantifications.

Quantifiers : Types

- THE UNIVERSAL QUANTIFIER
- THE EXISTENTIAL QUANTIFIER

The Universal Quantifier

Definition

The *universal quantification* of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain."

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier.** We read $\forall x P(x)$ as "for all $x P(x)$" or "for every $x P(x)$." An element for which $P(x)$ is false is called a **counterexample** of $\forall x P(x)$.

The Universal Quantifier : True/False?

Statement	When True?	When False?
$\forall x\, P(x)$	P(x) is true for every x	There is an x for which P(x) is false.

P(x) = "Student x knows C programming."

Domain : This Class!

When true?

If every student knows programming

When false?

If at least one student does not know

The Existential Quantifier

Definition

The *existential quantification* of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$."

We use the notation $\exists x\, P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

The Existential Quantifier : True/False?

Statement	When True?	When False?
$\exists x\ P(x)$	There is an x for which P(x) is true.	P(x) is false for every x.

P(x) = "Student **x** knows C programming."

Domain : This Class!

When true?

If at least one student knows programming

When false?

If every student know

Summarize

TABLE 1 Quantifiers.

Statement	*When True?*	*When False?*
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus. For example, $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$. In other words, it means $(\forall x P(x)) \vee Q(x)$ rather than $\forall x (P(x) \vee Q(x))$.

Quantifiers : Example

Let Q(x) be the statement “x < 2.” What is the truth value of the quantification ∀x Q(x), where the domain consists of all real numbers?

Solution : Q(x) is not true for every real number x, because, for instance, Q(3) is false. That is, x = 3 is a counter example for the statement ∀x Q(x). Thus ∀x Q(x) is false.

Let P(x) denote the statement “x > 3”. What is the truth value of the quantification ∃x P (x), where the domain consists of all real numbers?

Solution : Because “ x > 3” is sometimes true—for instance, when x = 4, the existential quantification of P(x), which is ∃x P(x), is true.

Negating Quantified Expressions

P(x) = "Student x in your class has taken a course in calculus."

- $\forall x\ P(x)$ = "Every student in your class has taken a course in calculus."
- $\exists x\ P(x)$ = "At least one student in this class has taken a course in calculus."

- $\forall x\ \neg P(x)$
- $\exists x\ \neg P(x)$

- $\neg\ \forall x\ P(x)$
- $\neg\ \exists x\ P(x)$

Negating Quantified Expressions

TABLE 2 **De Morgan's Laws for Quantifiers.**

Negation	*Equivalent Statement*	*When Is Negation True?*	*When False?*
$\neg\exists x\, P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall x\, P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

Multivariable Predicates

M(X, Y) : **X** has emailed **Y**

- $\forall X \; \forall Y \; M(X, Y)$
- $\forall X \; \exists Y \; M(X, Y)$
- $\exists X \; \forall Y \; M(X, Y)$
- $\exists X \; \exists Y \; M(X, Y)$

Multi Variable Quantifier Negation

M(X, Y) : **X** has emailed **Y**

• $\neg \forall X \forall Y M(X, Y)$	$\equiv \exists X \exists Y \neg M(X, Y)$
• $\neg \forall X \exists Y M(X, Y)$	$\equiv \exists X \forall Y \neg M(X, Y)$
• $\neg \exists X \forall Y M(X, Y)$	$\equiv \forall X \exists Y \neg M(X, Y)$
• $\neg \exists X \exists Y M(X, Y)$	$\equiv \forall X \forall Y \neg M(X, Y)$

Multi Variable Quantifier Negation

De-Morgans Law

$\neg\forall x\ \forall y\ P(x,y) \wedge \mathrm{Q}(\mathrm{x},\mathrm{y})$ $\equiv \exists x\ \exists y\ \neg p(x,y) \vee \neg Q(x,y)$

Task

Let *P(x) be the statement "x can speak Russian" and let Q(x) be the statement "x knows the computer language* C++." Express each of these sentences in terms of *P(x), Q(x), quantifiers, and logical connectives. The domain* for quantifiers consists of all students at your school.

a) There is a student at your school who can speak Russian and who knows C++.

b) There is a student at your school who can speak Russian but who doesn't know C++.

c) Every student at your school either can speak Russian or knows C++.

d) No student at your school can speak Russian or knows C++.

Task

- Let *L(x, y) be the statement "x loves y," where the domain* for both *x and y consists of all people in the world.* Use quantifiers to express each of these statements.
- **a) Everybody loves Jerry.**
- **b) Everybody loves somebody.**
- **c) There is somebody whom everybody loves.**
- **d) Nobody loves everybody.**

Rules of Inference

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \to q$ $\therefore q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \to p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

Example from Book!!!

Rules of Inference for Quantified Statements

TABLE 2 Rules of Inference for Quantified Statements.

Rule of Inference	*Name*
$\forall x P(x)$ $\therefore P(c)$	Universal instantiation
$P(c)$ for an arbitrary c $\therefore \forall x P(x)$	Universal generalization
$\exists x P(x)$ $\therefore P(c)$ for some element c	Existential instantiation
$P(c)$ for some element c $\therefore \exists x P(x)$	Existential generalization

Example from Book!!!